ACM/IDS 104 APPLIED LINEAR ALGEBRA
PROBLEM SET 1: SOLUTIONS

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Problem 1. (10 points) Matrix Multiplication
Let $A \in \mathbb{M}_{m\times p}$ and $B \in \mathbb{M}_{p\times n}$ be $m \times p$ and $p \times n$ matrices respectively. Then, by definition, the (i, j) entry of $C = AB$ is the *dot product* of the ith row of A and the jth column of B:

$$c_{ij} = a^i \cdot b_j = \sum_{k=1}^{p} a_{ik} b_{kj}. \tag{1}$$

Interestingly, we can also compute AB by multiplying columns of A by rows of B. Show that

$$AB = \sum_{k=1}^{p} a_k b^k, \tag{2}$$

where each term $a_k b^k$ is a matrix of size $m \times n$. In particular, if B is a column vector $B = x \in \mathbb{R}^p$, then

$$Ax = \sum_{k=1}^{p} x_k a_k. \tag{3}$$

is a linear combination of columns $a_1, \dots, a_p$ of A with coefficients $x_1, \dots, x_p$ (components of x). We will use this important interpretation of matrix multiplication in lectures.

Solution:
Let's show that the (i, j) entry $(\sum_{k=1}^{p} a_k b^k)_{ij}$ of matrix $\sum_{k=1}^{p} a_k b^k$ coincides with $c_{ij} = \sum_{k=1}^{p} a_{ik} b_{kj}$. This would justify (2). By definition of matrix addition,

$$\left(\sum_{k=1}^{p} a_k b^k\right)_{ij} = \sum_{k=1}^{p} (a_k b^k)_{ij}. \tag{4}$$

Let's look at the (i, j) entry $(a_k b^k)_{ij}$ of the kth matrix $a_k b^k$. It is the ith entry of the column vector a_k multiplied by the jth entry of the row vector b^k, that is $(a_k b^k)_{ij} = a_{ik} b_{kj}$. Therefore,

$$\left(\sum_{k=1}^{p} a_k b^k\right)_{ij} = \sum_{k=1}^{p} a_{ik} b_{kj} = c_{ij}. \tag{5}$$

Problem 2. (10 points) Matrix Identities
It this problem, we will explore how familiar properties of numbers and functions look for matrices.

(a) (3 points) If a and b are non-zero numbers, then

$$\frac{1}{a} - \frac{1}{b} = \frac{b-a}{ab}. \tag{6}$$

Let A and B be nonsingular matrices of the same size. Find an analog of (6) for $A^{-1} - B^{-1}$. That is, formulate an intelligent guess for

$$A^{-1} - B^{-1} = \dots \tag{7}$$

and prove it.

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(b) (3 points) Let $A(t)$ be a family of nonsingular matrices depending smoothly on the real parameter $t \in \mathbb{R}$. Find and prove a formula for the derivative of the inverse $(A^{-1}(t))'$ in terms of the derivative $A'(t)$ and the inverse $A^{-1}(t)$. Check that the derived formula works in a special case, where $A(t)$ is a family of 1-by-1 matrices, that is, a family of functions.

(c) (4 points) Let $A(t)$ be a family of square matrices depending smoothly on the real parameter $t \in \mathbb{R}$. Find and prove a formula for the derivative of $A^2(t)$.

Solution:

(a) Looking at (6), the first that comes into mind is

$$A^{-1} - B^{-1} = A^{-1}(B - A)B^{-1}. \tag{8}$$

It is easy to prove that this is indeed the case:

$$\begin{aligned} &A^{-1} - B^{-1} = A^{-1}(B - A)B^{-1} \iff \\ &I - AB^{-1} = (B - A)B^{-1} \iff \\ &B - A = B - A. \end{aligned} \tag{9}$$

Alternatively, it can be shown that

$$A^{-1} - B^{-1} = B^{-1}(B - A)A^{-1}. \tag{10}$$

(b) By definition,

$$(A^{-1}(t))' = \lim_{h \to 0} \frac{A^{-1}(t+h) - A^{-1}(t)}{h}. \tag{11}$$

Now lets use (8):

$$\begin{aligned} (A^{-1}(t))' &= \lim_{h \to 0} \frac{A^{-1}(t+h) - A^{-1}(t)}{h} \\ &= \lim_{h \to 0} \frac{A^{-1}(t+h)(A(t) - A(t+h))A^{-1}(t)}{h} = -A^{-1}(t)A'(t)A^{-1}(t). \end{aligned} \tag{12}$$

For functions, we know that

$$\left(\frac{1}{f(t)}\right)' = -\frac{f'(t)}{f^2(t)}. \tag{13}$$

If $A(t)$ is a function, then (12) reduces to (13).

(c) By definition,

$$(A^2(t))' = \lim_{h \to 0} \frac{A^2(t+h) - A^2(t)}{h}. \tag{14}$$

It may be not clear how to proceed from here. Note, that $A^2 - B^2 \neq (A-B)(A+B)$, unless $AB = BA$. But we know, that whatever formula we derive, it should reduce to the well known formula for functions if $A(t) = f(t)$ is a function (1-by-1 matrix):

$$(f^2(t))' = 2f(t)f'(t). \tag{15}$$

This leads to an intelligent guess for the matrix case:

$$(A^2(t))' = A(t)A'(t) + A'(t)A(t). \tag{16}$$

Having this in mind suggests how to proceed in (14):

$$\begin{aligned} (A^2(t))' &= \lim_{h \to 0} \frac{A^2(t+h) - A^2(t)}{h} \\ &= \lim_{h \to 0} \frac{A(t+h)(A(t+h) - A(t)) + (A(t+h) - A(t))A(t)}{h} \\ &= A(t)A'(t) + A'(t)A(t). \end{aligned} \tag{17}$$

Problem 3. (10 points) Permuted LU Decomposition
Let A_n be the $n \times n$ coefficient matrix of a linear system which we obtained in Lecture 1 by discretizing the Poisson equation. That is, let A_n be the $n \times n$ tridiagonal matrix with all 2's along the main diagonal, and all -1's along the sub- and super-diagonals. Find a permuted LU decomposition of matrix A_n, that is $P_n A_n = L_n U_n$, where P_n is a permutation matrix, L_n is special lower triangular, and U_n is upper triangular with non-zero diagonal elements.

Hint: It may be helpful to solve Practice Problems 1A and 2A before solving this problem.

Solution: Matrix A_n is nonsingular. Let's perform the elementary row operations to reduce A_n to an upper triangular matrix U_n with non-zero diagonal elements:

$$A_n = \begin{bmatrix} 2 & -1 & 0 & \cdots & 0 & 0 \\ -1 & 2 & -1 & \ddots & & \vdots \\ 0 & -1 & 2 & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -1 & 0 \\ 0 & & \ddots & -1 & 2 & -1 \\ 0 & \cdots & 0 & 0 & -1 & 2 \end{bmatrix} \xrightarrow{a^2 \to a^2 + \frac{1}{2}a^1} \begin{bmatrix} 2 & -1 & 0 & \cdots & 0 & 0 \\ 0 & \frac{3}{2} & -1 & \ddots & & \vdots \\ 0 & -1 & 2 & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -1 & 0 \\ 0 & & \ddots & -1 & 2 & -1 \\ 0 & \cdots & 0 & 0 & -1 & 2 \end{bmatrix} \xrightarrow{a^3 \to a^3 + \frac{2}{3}a^2}$$

$$= \begin{bmatrix} 2 & -1 & 0 & \cdots & 0 & 0 \\ 0 & \frac{3}{2} & -1 & \ddots & & \vdots \\ 0 & 0 & \frac{4}{3} & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -1 & 0 \\ 0 & & \ddots & -1 & 2 & -1 \\ 0 & \cdots & 0 & 0 & -1 & 2 \end{bmatrix} \xrightarrow{a^4 \to a^4 + \frac{3}{4}a^3} \cdots \xrightarrow{a^n \to a^n + \frac{n-1}{n}a^{n-1}} \begin{bmatrix} 2 & -1 & 0 & \cdots & 0 & 0 \\ 0 & \frac{3}{2} & -1 & \ddots & & \vdots \\ 0 & 0 & \frac{4}{3} & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -1 & 0 \\ 0 & & \ddots & 0 & \frac{n}{n-1} & -1 \\ 0 & \cdots & 0 & 0 & 0 & \frac{n+1}{n} \end{bmatrix} = U_n \tag{18}$$

Equivalently, in matrix form (using how row operations are realized by matrix multiplications):

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \ddots & \vdots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 1 & 0 \\ 0 & 0 & \cdots & 0 & \frac{n-1}{n} & 1 \end{bmatrix}}_{E_{n,n-1}^{((n-1)/n)}, a^n \to a^n + \frac{n-1}{n}a^{n-1}} \cdots \underbrace{\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & \frac{2}{3} & 1 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix}}_{E_{32}^{(2/3)}, a^3 \to a^3 + \frac{2}{3}a^2} \underbrace{\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ \frac{1}{2} & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix}}_{E_{21}^{(1/2)}, a^2 \to a^2 + \frac{1}{2}a^1} A_n = U_n \tag{19}$$

We have not performed any swapping of rows (type II operations). Therefore, the permutation matrix is the $n \times n$ identity matrix $P_n = I_n$. The lower triangular matrix L_n is

$$
\begin{aligned}
L_n &= \left(\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \ddots & \vdots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 1 & 0 \\ 0 & 0 & \cdots & 0 & \frac{n-1}{n} & 1 \end{bmatrix} \cdots \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & \frac{2}{3} & 1 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ \frac{1}{2} & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix} \right)^{-1} \\
&= \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ \frac{1}{2} & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & \frac{2}{3} & 1 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix}^{-1} \cdots \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \ddots & \vdots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 1 & 0 \\ 0 & 0 & \cdots & 0 & \frac{n-1}{n} & 1 \end{bmatrix}^{-1} \\
&= \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ -\frac{1}{2} & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & -\frac{2}{3} & 1 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix} \cdots \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \ddots & \vdots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 1 & 0 \\ 0 & 0 & \cdots & 0 & -\frac{n-1}{n} & 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ -\frac{1}{2} & 1 & 0 & \cdots & 0 \\ 0 & -\frac{2}{3} & 1 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix} \cdots \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \ddots & \vdots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 1 & 0 \\ 0 & 0 & \cdots & 0 & -\frac{n-1}{n} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ -\frac{1}{2} & 1 & 0 & \cdots & 0 \\ 0 & -\frac{2}{3} & 1 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ 0 & 0 & \cdots & -\frac{n-1}{n} & 1 \end{bmatrix}
\end{aligned}
\tag{20}
$$

So, the permuted LU decomposition of A_n looks as follows:

$$
\underbrace{I_n}_{P} \underbrace{\begin{bmatrix} 2 & -1 & 0 & \cdots & 0 & 0 \\ -1 & 2 & -1 & \ddots & & \vdots \\ 0 & -1 & 2 & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -1 & 0 \\ 0 & & \ddots & -1 & 2 & -1 \\ 0 & \cdots & 0 & 0 & -1 & 2 \end{bmatrix}}_{A_n} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ -\frac{1}{2} & 1 & 0 & \cdots & 0 \\ 0 & -\frac{2}{3} & 1 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ 0 & 0 & \cdots & -\frac{n-1}{n} & 1 \end{bmatrix}}_{L_n} \underbrace{\begin{bmatrix} 2 & -1 & 0 & \cdots & 0 & 0 \\ 0 & \frac{3}{2} & -1 & \ddots & & \vdots \\ 0 & 0 & \frac{4}{3} & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -1 & 0 \\ 0 & & \ddots & 0 & \frac{n}{n-1} & -1 \\ 0 & \cdots & 0 & 0 & 0 & \frac{n+1}{n} \end{bmatrix}}_{U_n}
\tag{21}
$$

Problem 4. (10 points) Types of Matrices

In this course, we will discuss many different types of matrices which occur in various applications. In this problem, we will briefly explore the properties of orthogonal and skew-symmetric matrices.

The Gaussian elimination method results in the permuted LU decomposition, $PA = LU$, which is fundamental for solving linear systems. Another important factorization of a nonsingular matrix is the so-called QR factorization (which we will discuss in lectures), which can also be used for solving square

linear systems (this approach is more computationally expensive, less prone to inaccuracies, and more numerically stable). The QR factorization is based on the notion of an *orthogonal* matrix. A square matrix A is called orthogonal if its inverse coincides with its transpose: $A^T = A^{-1}$.

(a) (3 points) Show that any permutation matrix P is orthogonal.
(b) (3 points) Is an orthogonal matrix necessarily a permutation matrix?

Other important classes of matrices are *symmetric* and *skew-symmetric* matrices. A square matrix A is symmetric if $A^T = A$, and is skew-symmetric if $A^T = -A$.

(c) (4 points) Show that every square matrix A can be written as a sum of a symmetric matrix S and a skew-symmetric matrix J:

$$A = S + J, \quad S^T = S, \quad J^T = -J. \tag{22}$$

Prove that this decomposition is unique and find the expressions for S and J in terms of A.

Solution:

(a) A matrix Q is orthogonal if $QQ^T = I$. We thus want to show that $PP^T = I$. Let's compute the (i, j) entry of PP^T:

$$(PP^T)_{ij} = \sum_{k=1}^{n} P_{ik}(P^T)_{kj} = \sum_{k=1}^{n} P_{ik}P_{jk} = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases} \tag{23}$$

The last equality is because every row and column in a permutation matrix contains exactly one 1 and all the other elements are 0. For the identity matrix:

$$I_{ij} = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases} \tag{24}$$

We therefore have $PP^T = I$ and P is orthogonal.

(b) No. For example, the matrix

$$R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \tag{25}$$

is not a permutation matrix for any $\theta \neq 2\pi n$, $n \in \mathbb{Z}$ ($n = 0, \pm 1, \pm 2, \ldots$), but

$$RR^T = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I. \tag{26}$$

(c) For any square matrix A,

$$A = \underbrace{\frac{A + A^T}{2}}_{S} + \underbrace{\frac{A - A^T}{2}}_{J}. \tag{27}$$

It is easy to check that S is symmetric and J is skew-symmetric.

To prove that the decomposition of a matrix into symmetric and skew-symmetric matrices is unique, suppose it is not:

$$A = S_1 + J_1 = S_2 + J_2, \tag{28}$$

where S_1 and S_2 are symmetric and J_1 and J_2 are skew-symmetric. But then

$$S_1 - S_2 = J_2 - J_1. \tag{29}$$

However, $S_1 - S_2$ is symmetric and $J_2 - J_1$ is skew-symmetric, which implies that $S_1 - S_2 = J_2 - J_1 = 0$, since only the zero matrix is both symmetric and skew-symmetric. Hence, $S_1 = S_2$ and $J_1 = J_2$, and the decomposition is unique.

Problem 5. (10 points) Solving Linear Systems
Let B be the following $n \times n$ matrix:

$$B = \begin{bmatrix} 1 & 2 & \dots & n \\ n+1 & n+2 & \dots & 2n \\ \vdots & \vdots & & \vdots \\ n^2-n+1 & n^2-n+2 & \dots & n^2 \end{bmatrix} \tag{30}$$

(a) (5 points) Find the rank of B and complete Problem 5a in `PS1.mlx`.
Remark: You don't need MATLAB to find the answer, but you can use it for making the right guess. We have provided a way of checking your answer in Problem 5a of `PS1.mlx`. In your solutions, please explain why your answer is correct for any n, and not only the values explored in MATLAB.

(b) (5 points) Complete Problem 5b in `PS1.mlx`. Remember to report the non-zero components in your solutions.

Solution:

Part (a): We know (a corollary of the fundamental theorem, see lecture 4) that the rank of a matrix is the dimension of its "row space," i.e. a subspace spanned by its rows. Let a^i be the ith row of A. The first two rows form a basis of $\text{span}(a^1, \dots, a^n)$. Indeed, they are linearly independent and

$$a^k = a^1 + (k-1)(a^2 - a^1), \quad k = 1, \dots, n. \tag{31}$$

See `PS1solutions.mlx` (uploaded to the Piazza page).

Part (b): See `PS1solutions.mlx` (uploaded to the Piazza page).

Acknowledgements. A huge thanks to Pavlos Stavrinides for editing my clumsy MATLAB m-files and converting them into neat live scripts!

ACM/IDS 104 - Problem Set 1- MATLAB Solutions

Before writing your MATLAB code, it is always good practice to get rid of any leftover variables and figures from previous scripts.

```
clc; clear; close all;
```

NOTE: As this is the first problem set (and many of you might be unfamiliar with MATLAB) we will provide some helper code. As the term progresses (and you become more experienced) we will omit this.

Problem 5 (10 points) Solving Linear Systems

We have the matrix:

$$B = \begin{pmatrix} 1 & 2 & \cdots & n \\ n+1 & n+2 & \cdots & 2n \\ \vdots & \vdots & & \vdots \\ n^2-n+1 & n^2-n+2 & \cdots & n^2 \end{pmatrix}$$

Part (a) (5 points)

In this part, your task is to find $\text{rank}(B)$. As mentioned in the problem set, MATLAB is not needed to obtain the answer. However, we can use MATLAB to make a right guess and check our answer. To do this, we first need to construct matrix B in MATLAB:

NOTE: Although you can check your answer here, you still need to justify and show your reasoning to obtain full credit :)

```
n = 100; % set n as specified in Part (b)
B = 1 : n;
for i = 2 : n
    % TODO (one-liner)
    B(i, :) = B(i - 1, :) + n;
end
r = rank(B); % check your answer here
```

Part (b) (5 points)

Set $n = 100$ and consider the system of linear equations $Bx = c$ where $c = (1 \quad 2 \quad \cdots \quad n)^T$. Find a solution x such that its first $[n - \text{rank}(B)]$ components are zero. What are the non-zero components of x?

HINT: The backslash operator $B\backslash c$ issues a warning if B is nearly singular and raises an error condition if it detects exact singularity. In that case, use `pinv(B)*c` for finding a particular solution of $Bx = c$. The function `pinv(B)` returns the "pseudoinverse" of B (will discuss the Moore-Penrose pseudoinverse in lecture 16). Also, the following built-in function may be useful: `null`.

```
%{
Let us start by defining the column vector c as specified above.
Remember that in MATLAB we can use ' to transpose a vector.
%}
```

```
c = [1 : n]';

%{
Now, we obtain a particular solution as described above
%}
x_0 = pinv(B) * c;

%{
Use null(B) to define a matrix whose columns form an orthonormal
basis in the vector space of all solution of the homogeneous
system Bx=0.
%}
V = null(B);

%{
Use the rank, r, found in part (a) to define k = n - rank(B).
This is the number of free variables / dimension of the vector
space.
%}
k = n - r;
```

This is a good point to review what we have done so far. Recall that the desired solution of the system is:

$$x = x_0 + V\alpha \quad (\star)$$

where α is a $k \times 1$ vector. We can obtain α by solving the system:

$$x_0 + V\alpha = 0 \quad (\star\star)$$

```
%{
Find alpha by solving the described system (**).
-> Hint1: Remeber that alpha is a k*1 vector. Hence, you need to
restrict the sizes of x_0 and V
-> Hint2: Use backslash
%}
alpha = - V(1 : k, :) \ x_0(1 : k);

%{
Finally, put everything together and find x using (*)
%}
x = x_0 + V*alpha;
disp(x);
```

```
   -0.0000
    0.0000
    0.0000
    0.0000
   -0.0000
    0.0000
   -0.0000
    0.0000
    0.0000
   -0.0000
```

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```
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    0.0000
    0.0000
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   -0.0000
   -0.0000
   -0.0000
    0.0000
    0.0100
```

```
%{
Now, let us see how x compares to the actual solution.
Un-comment the following 2 lines of code once you reach this part.
%}
error = norm(B*x - c);
disp(error);
```

```
4.5386e-12
```

Don't forget to report the non-zero components of x!